Binomial Hypothesis Testing

Questions 1–20

Question 1. Describe what is meant by the significance level.

Question 2. In each case, find the p-value and determine whether you reject H_0 or fail to reject H_0

a) $H_0 : p = 0.7$, $H_1 : p < 0.7$
$X \sim B(20, 0.7)$
Observed 10, 5% significance level

b) $H_0 : p = 0.3$, $H_1 : p < 0.3$
$X \sim B(30, 0.3)$
Observed 5, 5% significance level

c) $H_0 : p = 0.6$, $H_1 : p < 0.6$
$X \sim B(27, 0.6)$
Observed 12, 10% significance level

d) $H_0 : p = 0.9$, $H_1 : p < 0.9$
$X \sim B(52, 0.9)$
Observed 40, 1% significance level

Question 3. In each case, find the p-value and determine whether you reject H_0 or fail to reject H_0

a) $H_0 : p = 0.4$, $H_1 : p > 0.4$
$X \sim B(15, 0.4)$
Observed 9, 5% significance level

b) $H_0 : p = 0.85$, $H_1 : p > 0.85$
$X \sim B(45, 0.85)$
Observed 41, 5% significance level

c) $H_0 : p = 0.55$, $H_1 : p > 0.55$
$X \sim B(36, 0.55)$
Observed 26, 10% significance level

d) $H_0 : p = 0.12$, $H_1 : p > 0.12$
$X \sim B(90, 0.12)$
Observed 18, 1% significance level

Question 4. In each case, find the p-value and determine whether you reject H_0 or fail to reject H_0

a) $H_0 : p = 0.8$, $H_1 : p \neq 0.8$
$X \sim B(25, 0.8)$
Observed 15, 5% significance level

b) $H_0 : p = 0.72$, $H_1 : p \neq 0.72$
$X \sim B(33, 0.72)$
Observed 29, 5% significance level

c) $H_0 : p = 0.38$, $H_1 : p \neq 0.38$
$X \sim B(67, 0.38)$
Observed 33, 10% significance level

d) $H_0 : p = 0.05$, $H_1 : p \neq 0.05$
$X \sim B(180, 0.05)$
Observed 3, 1% significance level

Question 5. Describe what is meant by:

a) the critical region,

b) the acceptance region.

Question 6. In each case, find the critical region and write down the actual significance level.

a) $H_0 : p = 0.5$, $H_1 : p < 0.5$
$X \sim B(18, 0.5)$
5% significance level

b) $H_0 : p = 0.55$, $H_1 : p < 0.55$
$X \sim B(38, 0.55)$
5% significance level

c) $H_0 : p = 0.35$, $H_1 : p < 0.35$
$X \sim B(29, 0.35)$
10% significance level

d) $H_0 : p = 0.86$, $H_1 : p < 0.86$
$X \sim B(65, 0.86)$
1% significance level

Question 7. In each case, find the critical region and write down the actual significance level.

a) $H_0 : p = 0.2$, $H_1 : p > 0.2$
$X \sim B(40, 0.2)$
5% significance level

b) $H_0 : p = 0.33$, $H_1 : p > 0.33$
$X \sim B(49, 0.33)$
5% significance level

c) $H_0 : p = 0.11$, $H_1 : p > 0.11$
$X \sim B(65, 0.11)$
10% significance level

d) $H_0 : p = 0.74$, $H_1 : p > 0.74$
$X \sim B(125, 0.74)$
1% significance level

Question 8. In each case, find the critical region and write down the actual significance level.

a) $H_0 : p = 0.4$, $H_1 : p \neq 0.4$
$X \sim B(60, 0.4)$
5% significance level

b) $H_0 : p = 0.9$, $H_1 : p \neq 0.9$
$X \sim B(44, 0.9)$
5% significance level

c) $H_0 : p = 0.28$, $H_1 : p \neq 0.28$
$X \sim B(55, 0.28)$
10% significance level

d) $H_0 : p = 0.09$, $H_1 : p \neq 0.09$
$X \sim B(135, 0.09)$
1% significance level

Question 9. Fran is a florist. She regularly makes the most sales the day before Mother's Day each year, and over several years 25% of customers are found to purchase a chrysanthemum bouquet on that day. Fran decides to make the chrysanthemum bouquets larger for the following year and consequently they become more expensive. She believes this will result in fewer bouquets of this variety being sold. The following year, in a random sample of 34 customers, 5 customers bought a chrysanthemum bouquet. Fran carries out a hypothesis test, at the 10% significance level, to test her belief. Fran's hypothesis test is shown below.

Let X be the number of customers purchasing a chrysanthemum bouquet.

$\mathrm{H}_0 : p = 0.25$

$\mathrm{H}_1 : p \leq 0.25$

$X \sim \mathrm{B}(34, 0.25)$

$\mathrm{P}(X = 5) = 0.0647$

$0.0647 > 0.1$ *so we fail to reject* H_0

There is insufficient evidence to suggest that the proportion of customers purchasing a chrysanthemum bouquet has decreased.

a) Identify the first two errors made by Fran in her hypothesis test.

b) Correct her test.

Question 10. Chaisai is playing a game of chance that is designed to have a probability of winning of $\frac{1}{5}$. Chaisai believes he has found a way to improve his chances and wishes to conduct a hypothesis test to test this belief. He plays the game 45 more times and wins 15 times. Carry out a hypothesis test at the 5% significance level to test Chaisai's claim.

Question 11. The National Health and Medical Research Council (NHMRC) 2013 Australian Dietary Guidelines set out the recommended servings of fruit and vegetables to be eaten per day. In 2022, it was reported that 44.1% of adults met the fruit recommendation of 2 servings. Amy believes that this proportion is too high. She takes a random sample of 60 adults and 20 of them reported they met the fruit recommendation of 2 servings a day. Carry out a hypothesis test at the 5% significance level to investigate whether Amy's belief is supported.

Question 12. The UK employment rate for Sep-Nov 2023 was 75.8%. Daniel lives in Brighton and believes the employment rate there is significantly lower than the UK average. He takes a random sample of 110 adults between the ages of 16 to 64 and finds that 70 of them are employed. Carry out a hypothesis test at the 1% significance level to investigate whether Daniel's belief is supported.

Question 13. Energee lithium-ion batteries have a 77% chance of lasting longer than 20 hours on a single charge. Energee releases a new version of the battery which they hope will on average last longer. A random sample of 150 of the new batteries is taken and 126 last longer than 20 hours. Carry out a hypothesis test at the 5% level of significance to investigate Energee's claim about the new batteries.

Question 14. Thomas works for a company that designs and produces television screens. Thomas leads the team that develops the remote control, and it has been found that 4% of the remote controls produced have had a fault. A change is made to the manufacturing process and Thomas wants to determine if this has had any effect on changing the percentage that have a fault. A random sample of 60 brand new remote controls are checked and six are found to have the fault. Carry out a hypothesis test at the 5% significance level to investigate whether the new manufacturing process has changed the percentage that have a fault.

Question 15. It is known that 35% of roses grown by a horticulturist suffer from powdery mildew. The horticulturist separates the roses from their other plants and grows them in a large new greenhouse instead. They want to conduct a hypothesis test to determine whether this has had any change on the proportion that suffer from powdery mildew. A random sample of 300 roses from the new greenhouse are collected and 81 are found to suffer from powdery mildew. Test, at the 3% significance level, whether there is any evidence to suggest the proportion has changed.

Question 16. Chloe and Ajay are playing a board game with a single six-sided die. After several turns, Chloe believes the die is bias in favour of scoring a 1. They decide to pause the board game and roll the die 35 times to test this theory.

a) Write down suitable hypotheses to test Chloe's belief.
b) Using a 10% significance level, find the critical region for the test.
c) Of the 35 rolls, 1 appears seven times. Complete the test.

Question 17. The probability of having to wait longer than three hours at a hospital emergency unit is known to be 76%. Another doctor is employed in the hope that waiting times will be reduced. A hypothesis test is to be conducted to test this belief.

a) Write down the hypotheses that should be used to test whether there has been a reduction in the number of patients waiting longer than three hours.

b) A random sample of 70 patients is taken. Using a 1% level of significance, find the critical region for the test.

Of the 70 patients in the sample, 43 waited longer than three hours to be seen.

c) Complete the test.

Question 18. A large supermarket sells soup by the can or in packs of six. Of those customers who bought soup in 2022, it was recorded that 32% bought packs of six. During the following year, Becky, the manager of the supermarket, believes that the percentage has changed. Using the 2023 records, she collects a random sample of 80 customers who bought cans of soup.

a) Write down the hypotheses that should be used to test Becky's belief.

b) Using a 5% significance level, find the critical region for a two-tailed test.

c) Find the actual significance level of a test based on your critical region.

Of the 80 customers who were selected in the sample, 45 bought cans of soup in packs of six.

d) Complete the test.

Question 19. Aaron Farms grows grain for stockfeed. Richard, a farmer, receives bags of grain from Aaron Farms and knows from experience that 5% will be spoilt. Richard decides to test whether another supplier, Baker Farms, will deliver fewer spoilt bags of grain. Richard makes several orders with Baker Farms and takes a random sample of 550 bags of grain.

a) Write down suitable hypotheses to test whether the number of bags of spoilt grain has been reduced.
b) Using a 10% significance level, find the critical region such a test.
c) Of the 550 bags of grain, 13 are found to be spoilt. Complete the test.

Question 20. John collects dinosaur models that are sold in sealed egg containers that the collector cannot see through. On average, 8% of the egg containers contain a limited edition gold dinosaur. John purchases 25 of the eggs. This may be considered to be a random sample.

a) Find the probability that
   i) exactly one of the eggs contains a gold dinosaur,
   ii) at most two eggs contain gold dinosaurs.
b) John buys 30 more eggs from a website. This may also be considered to be a random sample. John believes there is a greater chance of obtaining gold dinosaurs online. John decides to conduct a hypothesis test at the 5% level.
   i) Write down suitable hypotheses for the test.
   ii) Find the critical region for the test.
   iii) John opens the eggs to find there are four golden dinosaurs. Complete the test.